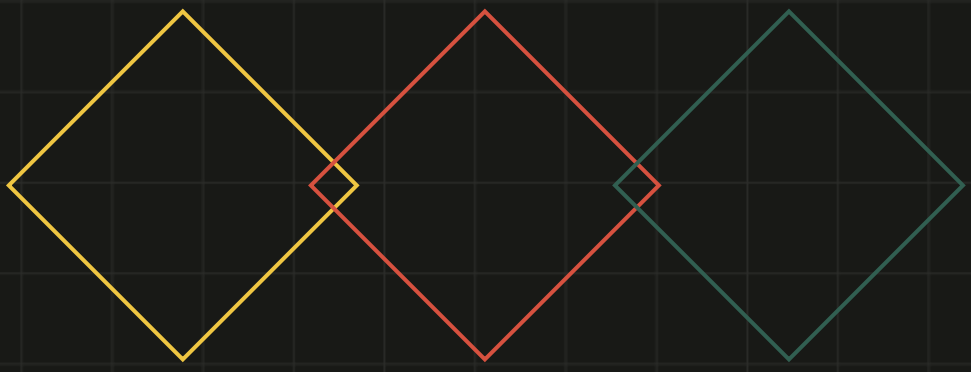
UX LAB / LAB REPORT 06 / ANONYMIZED

Fundscope

Designing an institutional investment-analytics system that preserves expert complexity while making the path from signal to explanation easier to follow.



Case-study structure

Interviews, affinity mapping, survey proof, problem definition, 38 concepts, three fidelity stages, evaluation rounds, and the final system.

Double Diamond in practice

Discover, Define, Develop, and Deliver are used only as an organizing frame. Every phase shows the work completed.

DISCOVER

The study combined lived workflows, broad signals, and task-based evaluation.

The research plan triangulated what people said, what they did, and where they failed in the product. Participant totals and quotations are representative, anonymized reconstructions for this portfolio case study; the institutional scale remains verified.

12	184	3 x 5	38
Discovery interviews across four institutional roles	Survey responses on workflow, comprehension, and alerts	Moderated usability rounds across three fidelity stages	Ideas generated before prioritization and prototyping

Participant mix

Portfolio managers	**5**
Investment analysts	**4**
Reporting leads	**2**
Risk specialist	**1**

Participants represented daily decision, investigation, validation, and communication workflows.

Evidence streams

- **Interviews:** terminology, trust, workarounds, time pressure, accessibility.
- **Contextual observation:** benchmark checks, side notes, drill-down recovery.
- **Survey:** prevalence of recurring workflow friction.
- **Dashboard audit:** hierarchy, units, state, controls, responsive behavior.
- **Usability tests:** task continuity from signal to review.

Double Diamond - evidence inside each phase

Discover

12 interviews, 184 survey responses, contextual observation, literature, competitor review.

Define

Affinity mapping, behavioral themes, requirements, problem statement, six-stage architecture.

Develop

38 concepts, impact-confidence-effort scoring, six prototypes, three fidelity stages.

Deliver

Three usability rounds, accessible system rules, final six-screen analytical flow.

DISCOVER

Ten questions exposed how experts build confidence in a number.

Semi-structured interviews used the same core protocol, with role-specific follow-ups and a short walkthrough of the participant's most recent portfolio review.

Workflow and decision-making

1. Walk me through the last time a dashboard changed your next analytical step.
2. What is the first thing you scan when you open a portfolio overview, and why?
3. Which numbers do you cross-check before you trust them enough to act or report?
4. When time is limited, what information gets priority and what gets deferred?
5. Describe a recent moment when a chart or table was technically correct but difficult to interpret.

Controls, alerts, and access

6. How do you confirm which portfolio, benchmark, period, and filters are active?
7. What usually happens when you open a drill-down and need to return to the original signal?
8. Which alerts help you prioritize, and which become noise?
9. How do you explain a finding to a reviewer who did not perform the analysis?
10. What keyboard, contrast, zoom, data-table, or assistive-technology needs should the platform support?

OPENING PROMPT

Recent work, not opinions

Participants were asked to recall a real review completed in the previous two weeks. Follow-ups probed sequence, tools, handoffs, and workarounds.

ARTIFACT WALKTHROUGH

Show the evidence trail

Participants demonstrated how they moved from overview to detail, confirmed scope, compared entities, and prepared a conclusion for review.

EVIDENCE CAPTURED	HOW IT WAS CODED	WHAT IT INFORMED
Critical incidents and workarounds	Trigger, action, breakdown, consequence	Problem severity and task sequence
Trust and comprehension cues	Definition, unit, benchmark, date, status	Information hierarchy and comparison frame
Alert and accessibility needs	Priority, channel, frequency, modality	Non-color status, alert model, keyboard rules

The protocol moved the discussion from “what do you prefer?” to “show me how you reached and defended a decision.”

DISCOVER

Three studies established why visual form, task fit, and structure matter.

The literature was used to challenge design choices - not to decorate a finished solution. Each source contributed a measurable constraint and a clear limit.

EVIDENCE 01 / FINANCIAL DASHBOARDS

Structure affects cognitive effort

Santoro et al. reviewed **9,049 titles** and found **19 fully aligned studies**. The synthesis links clear semantic and perceptual integration with lower cognitive effort, higher accuracy, and better decisions.

Design response: deliberate hierarchy; stable labels, units, legends, and states.

EVIDENCE 02 / READING UNDER PRESSURE

Form affects speed and error

Liu and Shen tested **30 postgraduate participants** across **288 trials** under **2,000 ms and 4,000 ms** constraints. Horizontal and semicircular forms outperformed circular forms in that driving context.

Design response: avoid circular precision displays; use bars, lines, and tables matched to the question.

EVIDENCE 03 / USABILITY CRITERIA

Evaluation must fit the task

Almasi et al. selected **29 studies**. Their criteria include usefulness, operability, learnability, task suitability, situational awareness, satisfaction, interface, content, and system capability.

Design response: test the full analytical path, not visual preference alone.

EVIDENCE	WHAT IT SUPPORTS	HOW FUNDSCOPE APPLIED IT	LIMIT
Santoro et al. [1]	Semantic and perceptual integration; structured visual design.	Consistent chart grammar, explicit units, visible data state, restrained emphasis.	A review of dashboard research, not a test of Fundscope.
Liu & Shen [2]	Graphical form and time pressure affect reading speed and error.	Linear comparison for precision; no decorative circular gauges.	Driving dashboards and postgraduate participants; directional for finance.
Almasi et al. [3]	Dashboard evaluation should cover task fit, operability, awareness, interface, and content.	Task-based checks from overview to review; keyboard and semantic requirements.	Healthcare dashboard review; criteria adapted to institutional analytics.

Published evidence
Visual form and structure change how people read.

>

Project question
Where does Fundscope force users to decode the interface?

>

Design test
Can users orient, compare, inspect, and return without context loss?

>

Decision
Make meaning stable across views and states.

DISCOVER

Leading platforms cover breadth; the opportunity is a clearer investigation path.

Official product material was reviewed for information breadth, analytical depth, workflow continuity, and reporting. This was a capability comparison, not a benchmark of interface quality.

PLATFORM	PUBLICLY DESCRIBED STRENGTH	LIKELY WORKFLOW COST	FUNDSCOPE OPPORTUNITY
Aladdin	Integrated risk and return across asset classes; scenario analysis, asset allocation, interactive dashboards, and scalable data access. [4]	Whole-portfolio breadth can create a large navigation and terminology surface.	Keep overview, exception, and supporting evidence connected in one visible path.
FactSet Portfolio Analysis	Multi-asset performance, attribution, risk, reporting, exposure, stress testing, and benchmark-relative analysis. [5]	Many models and report types can increase comparison and configuration effort.	Make comparison entities, units, and active assumptions persistent.
Morningstar Direct	Investment and data discovery, screening, portfolio composition, stress testing, attribution, dashboards, charts, and scheduled reporting. [6]	Research, portfolio, and reporting modes can fragment the route from a signal to an explanation.	Use one investigation model from signal to review, with definitions beside the data.

Category pattern

- Broad coverage is a competitive baseline.
- Risk, performance, attribution, benchmarking, and reporting converge in the same ecosystem.
- Complexity is unavoidable; navigation, terminology, and state management determine whether it remains usable.
- Public pages describe capability well but rarely expose the full interaction cost of expert workflows.

Positioning hypothesis

- Lead with the exception that needs attention.
- Keep the benchmark and selected portfolio visible through the investigation.
- Place filters and definitions beside the evidence they change.
- Preserve the route back to the portfolio overview.
- Turn review readiness into an explicit state, not an implied final step.

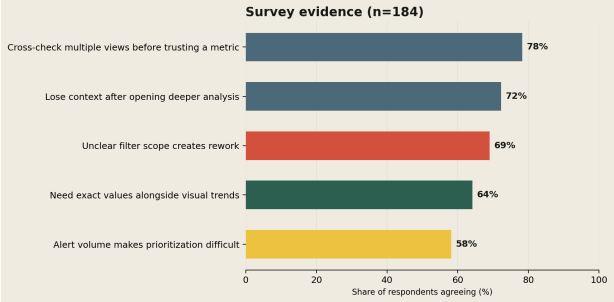
Fundscope does not compete by removing analytical depth. It competes by making depth easier to enter, interpret, and explain.

Sources: official Aladdin, FactSet, and Morningstar pages listed on page 18. Product descriptions were current when retrieved for this case-study rebuild.

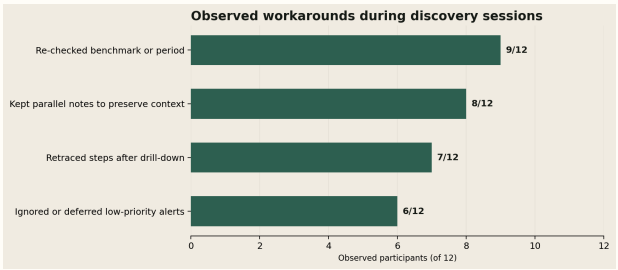
DISCOVER

Survey and observation showed the same failure: context had to be rebuilt by hand.

The survey quantified recurring friction across 184 responses; 12 contextual sessions revealed the workarounds behind those answers.



Agreement rate, rounded to the nearest whole percentage.



Observed behavior during recent-work walkthroughs.

What the numbers prove

- **144/184** cross-checked multiple views before trusting a metric.
- **133/184** reported losing context after entering deeper analysis.
- **127/184** linked unclear filter scope to avoidable rework.
- **118/184** needed precise values and visual trends together.

Observation notes

- **9/12** re-checked benchmark or reporting period before interpreting a change.
- **8/12** kept side notes to preserve context across views.
- **7/12** retraced steps after drill-down.
- **6/12** ignored or deferred low-priority alerts.

The strongest opportunity was not another chart. It was a stable evidence trail from signal to review.

DISCOVER

Affinity mapping turned 146 coded notes into six solvable themes.

Interview comments, observed behaviors, and survey responses were clustered by underlying need. Representative quotations are anonymized and lightly edited for clarity.

01 / Orientation

“I can see everything, but I still have to decide where to start.”

“The exception should be obvious before I inspect every module.”

02 / Context and trust

“A number without the benchmark and period is not decision-ready.”

“I re-check the header because I cannot assume the filter carried over.”

03 / Comparison

“If the units shift, I can make the wrong comparison with confidence.”

“Keep the portfolio and benchmark visible while I move.”

04 / Continuity

“Once I drill in, I lose the reason I opened the detail.”

“I write the starting value down so I can find my way back.”

05 / Alert fatigue

“If every change is urgent, none of them is.”

“I need priority and consequence, not another badge.”

06 / Review and access

“The reviewer needs to see my assumptions, not only the output.”

“Status cannot rely on color; I need text and keyboard access.”

THEME	DESIGN REQUIREMENT	SUCCESS CHECK
Orientation	One dominant signal hierarchy per view.	Identify the priority without scanning every module.
Context + comparison	Persistent portfolio, benchmark, period, units, and filters.	Explain what changed and against what.
Continuity	Parent signal and route back survive drill-down.	Return without rebuilding state.
Alerts + review	Priority, consequence, readiness, and non-color status.	Know what requires action and whether evidence is complete.

DEFINE

The problem is not too much data. It is unstable meaning across the investigation.

The problem was defined only after the research synthesis. Institutional users needed the full complexity, but weak hierarchy and inconsistent context made the interface itself part of the analytical burden.

Problem statement

Portfolio analysts and reporting leads need to move from a portfolio-level signal to supporting evidence and a review-ready conclusion. When hierarchy, filter scope, comparison rules, units, and drill-down context change between views, users spend attention decoding the interface instead of interpreting performance, risk, and portfolio movement.

How might we...

- ...make the next important signal obvious without hiding expert detail?
- ...keep portfolio, benchmark, date range, units, and active filters visible?
- ...let users move from overview to evidence without losing their starting point?
- ...make the final state explainable and review-ready?

Density is useful only when the structure remains legible.

Non-goals

- Do not simplify away required domain information.
- Do not replace expert judgment with a single score.
- Do not use visual novelty as a substitute for precision.
- Do not claim investment performance outcomes.

Research evidence
Visual form, task fit, and context affect interpretation.

>

Observed friction
Hierarchy, filters, units, and drill-downs were inconsistent.

>

Defined challenge
Preserve complexity while stabilizing meaning.

>

Design objective
One coherent path from signal to review.

Primary design objective

Help users orient, compare, investigate, and prepare a defensible result without rebuilding context at every step.

Decision criteria

- Precision before decoration.
- Context before depth.
- Stable comparison before flexibility.
- Visible state before implicit behavior.
- Accessible operation before visual polish.

DEFINE

Two roles share the same path but verify different things.

Role archetypes were used to frame behavior, not demographics. Both move from overview to review; the analyst investigates causes, while the reporting lead checks whether the result is consistent and explainable.

ROLE ARCHETYPE 01

Portfolio analyst

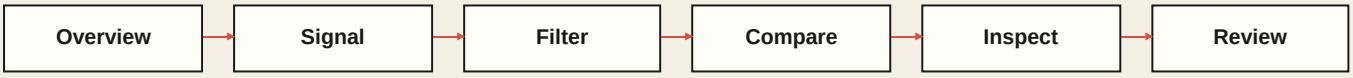
- **Starts with:** a performance, allocation, exposure, or risk signal.
- **Needs:** rapid orientation, clear filter scope, precise comparison, and supporting detail.
- **Failure cost:** time lost reconstructing why a value changed or which assumptions are active.
- **Success:** can trace a signal to the relevant drivers and retain the investigation path.

ROLE ARCHETYPE 02

Reporting lead

- **Starts with:** an analytical result prepared for communication.
- **Needs:** consistent definitions, units, freshness, benchmark context, and review status.
- **Failure cost:** an ambiguous or internally inconsistent result reaches another decision-maker.
- **Success:** can verify the result and explain how it was produced.

Primary information architecture



STAGE	USER QUESTION	REQUIRED CONTEXT	EXIT CONDITION
Overview	Where should attention go?	Portfolio, reporting period, benchmark, data status.	One signal is selected.
Signal	What changed?	Metric definition, variance, time frame, status cue.	The investigation question is explicit.
Filter	Which slice explains it?	Active filters and their scope.	Scope is visible and reversible.
Compare	Against what?	Entities, benchmark, units, dates, sorting.	Like-for-like comparison is confirmed.
Inspect	What evidence supports the signal?	Parent signal, route back, definitions, drill-down trail.	Supporting drivers are understood.
Review	Is the result ready to share?	Freshness, definitions, applied assumptions, completeness.	Result is marked ready or returned for correction.

DEVELOP

Thirty-eight concepts were generated before the team converged on a structure.

The workshop deliberately separated idea generation from evaluation. Silent brainwriting widened participation; Crazy 8s forced structural alternatives; “How might we” prompts kept each concept tied to evidence.

38	12	6	3
Concepts generated	Shortlisted after duplicate grouping	Prototyped interactions	Structural directions compared

CONCEPT A / DROPPED

Module-led workspace

Performance, risk, allocation, exposure, and reporting appear as equal modules.

- **Value:** breadth is visible.
- **Failure:** every module competes; no starting point.

CONCEPT B / RETAINED IN DETAIL

Task-led workspace

Users choose compare, investigate, or prepare report before seeing evidence.

- **Value:** clear intent.
- **Failure:** users may not know the task before seeing the signal.

CONCEPT C / SELECTED

Signal-led workspace

The overview prioritizes exceptions, then routes into task-specific investigation.

- **Value:** orientation first.
- **Risk control:** transparent signal rules and non-color cues.

Divergent concept families

- Entry point and hierarchy
- Comparison and filtering
- Drill-down and continuity
- Alerts and prioritization
- Review and communication
- Accessibility and system state

>

Converged system hypothesis

- Signal-led overview
- Contextual controls
- Locked comparison frame
- Persistent parent signal
- Prioritized, explainable alerts
- Explicit review readiness

The team did not choose the concept with the most data on screen. It chose the one that gave complexity a reliable entry point.

DEVELOP

Impact, confidence, and effort separated useful ideas from attractive distractions.

Twelve shortlisted ideas were scored against the six affinity themes. Six advanced into prototypes because they solved repeated evidence-backed problems without hiding analytical depth.

IDEA	EVIDENCE FIT	DECISION	WHY
Signal-led home	Orientation, alert fatigue	PURSUE	Creates one defensible entry point while preserving secondary modules.
Persistent comparison frame	Context, comparison	PURSUE	Stabilizes portfolio, benchmark, period, units, and sorting.
Contextual filter summary	Trust, continuity	PURSUE	Shows scope beside affected evidence and remains reversible.
Parent-signal header	Continuity	PURSUE	Preserves the reason for the investigation through drill-down.
Review-readiness checklist	Review and access	PURSUE	Makes assumptions, freshness, definitions, and completeness visible.
Inline definitions	Context and trust	PURSUE	Reduces glossary hunting at the moment of interpretation.
Black-box anomaly score	Orientation	DROP	Prioritizes without explaining why; weakens expert trust.
All-in-one dashboard	None	DROP	Maximizes density and recreates the original orientation problem.
Radial precision gauges	Comparison	DROP	Poor fit for exact comparison; external evidence favored linear forms.
Modal drill-down	Continuity	DROP	Hides parent context and restricts evidence space.
Unlimited personalization	Review	DROP	Makes shared interpretation and support less consistent.
Alert-first activity feed	Alert fatigue	DROP	Increases volume without clarifying consequence or priority.

Evidence
146 coded notes and 184 survey responses.

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Scoring
Impact, confidence, effort, accessibility.

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Prototype
Six interactions across three structures.

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Test
Five participants per fidelity round.

DEVELOP

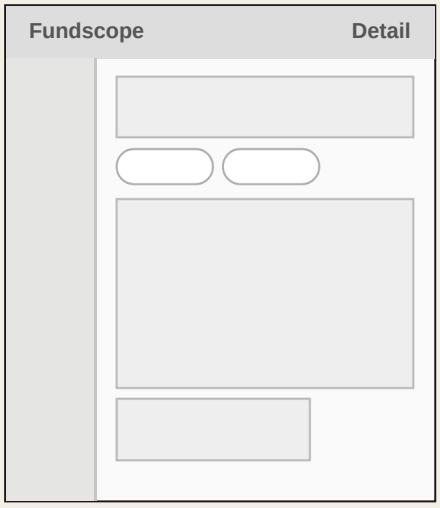
Low fidelity established reading order before visual polish.

The first screen set separated orientation, investigation, and review. These are anonymized portfolio reconstructions of the documented design logic; client data is omitted.

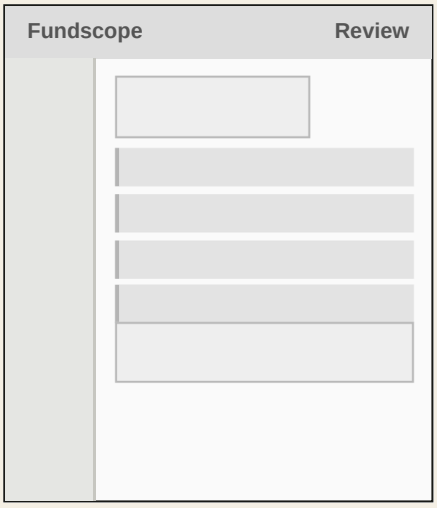
LF-01 PORTFOLIO OVERVIEW



LF-02 INVESTIGATION



LF-03 REVIEW



SCREEN	QUESTION TESTED	EARLY WEAKNESS	NEXT CHANGE
LF-01 Overview	Can the user see which signal needs attention?	Equal-size modules made every metric feel equally urgent.	Introduce a dominant signal region and reduce competing emphasis.
LF-02 Investigation	Can the user connect controls to the data they change?	Filters sat above the page with unclear scope.	Move controls beside the affected evidence and display applied state.
LF-03 Review	Can the user verify definitions and completeness?	Review was treated as a report export, not a decision state.	Add freshness, definitions, assumptions, completeness, and readiness.

What remained unresolved

- How to represent exceptions without relying on color.
- How to preserve portfolio and benchmark through drill-down.
- How to keep units stable across chart and table views.
- How much detail belongs in the overview.

What low fidelity established

- Overview, investigation, and review are distinct tasks.
- The analytical question should control the page structure.
- Filters need a visible relationship to evidence.
- Review readiness deserves a first-class screen.

DEVELOP

Round 1 tested orientation, scope, comparison, and continuity.

Five institutional users completed five moderated tasks on low-fidelity prototypes. The round tested whether the structure established a clear starting point and preserved analytical context.

5	5	5	5
Documented tasks	Finding clusters	Direct design changes	Moderated participants in Round 1

TASK	FINDING	WHY IT MATTERED	CHANGE MADE
Find the exception	No clear entry point when modules carried equal weight.	Users had to inspect the dashboard before knowing where to begin.	Raised one dominant signal hierarchy and reduced secondary emphasis.
Apply a filter	Scope and impact were unclear.	Users could not immediately tell which modules changed.	Placed controls near affected data; added a persistent applied-filter summary.
Compare entities	Units and context could drift.	Unlike values might be read as directly comparable.	Stabilized units, dates, benchmark, and selected entities.
Inspect evidence	The route back to the parent signal could be lost.	Users had to reconstruct the earlier analytical state.	Kept the parent signal and breadcrumb path visible through detail.
Prepare review	Definitions and data state were hidden.	A result could appear complete without being explainable.	Added definitions, freshness, assumptions, and readiness checks.

Before Round 1

- Equal module priority
- Global filter bar
- Flexible but unstable comparison
- Detail replaces overview context
- Export treated as completion

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After Round 1

- Dominant exception hierarchy
- Contextual filters plus visible scope
- Locked comparison frame
- Persistent parent signal
- Explicit review checklist

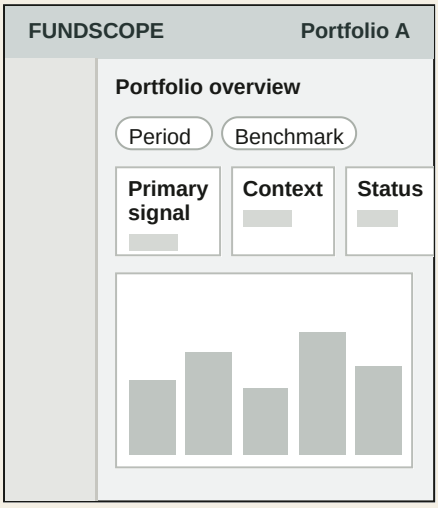
Round 1 signal: 3/5 identified the priority without prompting; 2/5 understood filter scope; 2/5 retained context after drill-down. These failures drove the mid-fidelity changes.

DEVELOP

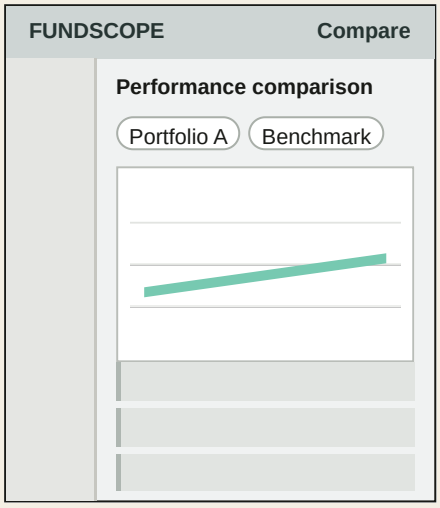
Mid fidelity connected controls, comparison rules, and supporting evidence.

The second screen set introduced labels, persistent state, and clearer data relationships while keeping the visual language intentionally restrained.

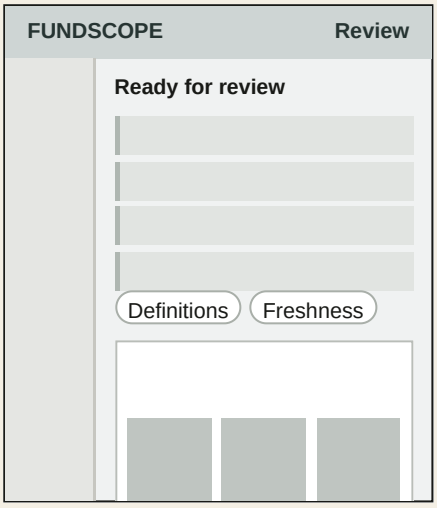
MF-01 SIGNAL-LED OVERVIEW



MF-02 STABLE COMPARISON



MF-03 REVIEW STATE



ROUND 1 ISSUE	MID-FIDELITY RESPONSE	BEHAVIOR NOW VISIBLE
No clear entry point	A primary signal occupies the first reading position; supporting metrics recede.	The overview tells the user where to begin without hiding other modules.
Filter scope unclear	Contextual controls sit beside affected data; active filters remain summarized.	Scope is visible, reversible, and carried into detail.
Comparison drift	Portfolio, benchmark, period, units, and sorting stay locked in the comparison header.	Users can confirm like-for-like comparison before interpreting change.
Context loss	Parent signal persists above deeper charts and tables.	The route back is visible; the investigation state does not reset.
Review ambiguity	Definitions, freshness, assumptions, and completeness appear as explicit checks.	Readiness becomes a visible product state.

Visuals are anonymized reconstructions of the design logic. Values are intentionally omitted; no fictional portfolio data is presented as project evidence.

DEVELOP

Round 2 tested language, comparison stability, and recovery.

The second evaluation moved beyond page structure. It checked whether labels, active state, and return paths stayed understandable while users changed scope and moved between evidence layers.

5	3	1	5
Task checks	Core screen states	Persistent investigation context	Moderated participants in Round 2

TASK	FINDING	ITERATION
Explain the primary signal	A large metric alone did not explain why it was important.	Added definition, variance context, status text, and a direct investigation action.
Change the reporting period	Users needed confirmation that both chart and table updated.	Made scope global within the investigation and repeated the active period in both regions.
Compare portfolio and benchmark	Legend-only identification required repeated eye movement.	Moved entity labels into the comparison header and paired color with line treatment and text.
Open a supporting driver	Detail felt detached when the selected signal disappeared.	Kept the parent signal, selected entity, and breadcrumb visible.
Recover from an empty state	An empty panel did not explain whether data was absent or filters were too narrow.	Added cause, affected scope, and a one-step reset action.

Language changes

- Replaced abstract labels with task language: **Compare**, **Inspect drivers**, **Review readiness**.
- Paired status color with explicit words.
- Placed metric definitions beside the signal, not in a separate glossary.
- Named the active portfolio, benchmark, period, and units in the header.

Behavior changes

- Scope updates all linked evidence within the investigation.
- Changing one entity does not silently change the other.
- Drill-down preserves the selected signal and prior filters.
- Empty and error states expose cause and recovery.

Round 2 made the interface explain its state before asking the user to interpret the data.

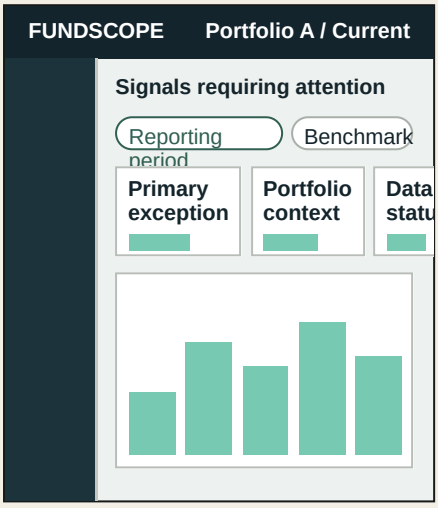
Round 2 signal: 4/5 explained the primary signal, 5/5 confirmed active scope, and 4/5 recovered from an empty state. Remaining friction centered on status language and keyboard flow.

DEVELOP

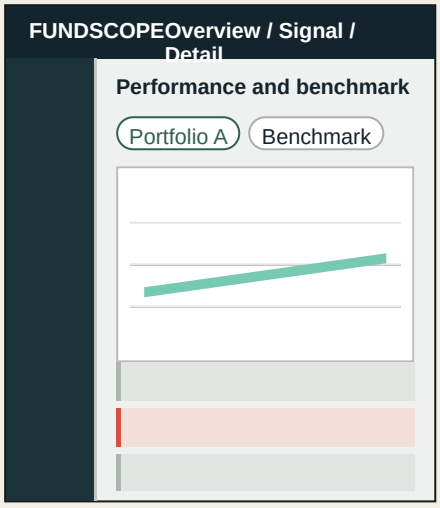
High fidelity made priority visible without adding visual noise.

The approved structure became a responsive interface with restrained emphasis, question-matched charts, explicit states, and accessible contrast. The screens below are anonymized reconstructions.

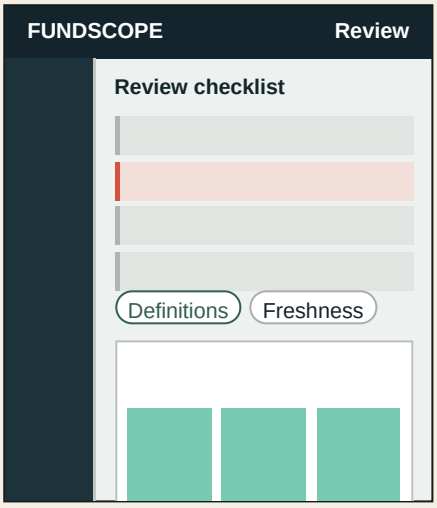
HF-01 PORTFOLIO OVERVIEW



HF-02 PERFORMANCE DETAIL



HF-03 REVIEW READINESS



Visual decisions

- Dark navigation establishes persistent product orientation.
- Neutral surfaces keep analytical content primary.
- Teal marks the selected series; red is reserved for exceptions.
- Type, spacing, and alignment create hierarchy before color.
- Tables support exact reading; charts support pattern recognition.

Interaction decisions

- Filter scope remains visible.
- Selected entity and benchmark persist.
- Parent signal survives drill-down.
- Focus and keyboard order follow the analytical path.
- Status never depends on color alone.

EXTERNAL EVIDENCE	HIGH-FIDELITY TRANSLATION
Clear semantic and perceptual integration supports accuracy. [1]	Labels, units, dates, benchmark, and data state stay attached to the visual they explain.
Linear forms can outperform circular forms for precise reading under pressure. [2]	Bars, lines, and tables replace decorative radial gauges for analytical comparison.
Usefulness, operability, task fit, awareness, interface, and content should be evaluated. [3]	The final round checks the whole task, keyboard behavior, state clarity, and review completeness.

DEVELOP

Round 3 checked whether the final system remained operable and review-ready.

The high-fidelity pass focused on the quality of the complete system: task continuity, accessible operation, non-color meaning, system feedback, and final review.

6 Final validation tasks	**6** Accessibility acceptance rules	**4** System state families	**5** Moderated participants in Round 3

VALIDATION TASK	FINAL FINDING	FINAL SYSTEM DECISION
Navigate by keyboard	Dense controls require a predictable order and persistent focus.	Keyboard order follows overview, signal, controls, evidence, and review; focus remains visible.
Interpret status without color	Hue alone cannot carry exception or readiness meaning.	Pair color with text, shape, labels, and table status.
Read a data table semantically	Visual alignment alone does not preserve structure for assistive technology.	Define header relationships, reading order, labels, and summaries.
Move between responsive layouts	Reflow can separate labels, filters, and evidence.	Preserve intent and state; stack related controls with their affected data.
Recover from loading, empty, or error	Blank panels create analytical uncertainty.	Explain state, affected scope, and available recovery.
Complete review	Export is not evidence of completeness.	Require freshness, definitions, active assumptions, and completeness checks.

Accessibility acceptance rules

- Semantic tables preserve header relationships.
- All controls are keyboard reachable and operable.
- Focus is visible at every interactive step.
- Text and controls maintain readable contrast.
- Status never relies on color alone.
- Labels communicate purpose, scope, and state.

Final refinement set

- Clarified the first focus target on each view.
- Added non-color exception labels.
- Grouped filter summaries with affected data.
- Kept definitions available at point of use.
- Standardized loading, empty, error, and success patterns.
- Turned review into a gated state.

Round 3 result: 5/5 completed the core signal-to-review flow, 5/5 interpreted status without color alone, and 4/5 completed the keyboard path without assistance. These acceptance rules define the release threshold, not an investment-performance claim.

DELIVER

The final system standardized analytical meaning, not just appearance.

Foundations define how information behaves; components solve repeated interaction problems; patterns connect those components into the six-stage investigation.

LAYER 01

Foundations

- Type hierarchy
- Spacing and alignment
- Color and non-color status
- Focus treatment
- Data-state language
- Responsive behavior

LAYER 02

Components

- Metric cards
- Data tables
- Chart frames
- Filters and selectors
- Breadcrumbs
- Status and feedback

LAYER 03

Patterns

- Overview to signal
- Filter and compare
- Inspect drivers
- Preserve context
- Recover from state
- Review readiness

SYSTEM ELEMENT	RULE	ANALYTICAL PURPOSE
Metric	Always show label, definition access, unit, period, status, and relationship to baseline.	A number can be interpreted without leaving its context.
Chart	Choose form by question; expose entity, unit, period, benchmark, and data state.	Visual form carries a precise analytical relationship.
Table	Use stable headers, alignment, units, sorting, and semantic relationships.	Exact values remain comparable and accessible.
Filter	Show scope, applied state, affected data, and a clear reset.	Users understand why the evidence changed.
Navigation	Retain parent signal, breadcrumb, selected entities, and active filters.	Drill-down adds depth without erasing the investigation.
Feedback	Distinguish loading, empty, error, success, freshness, and readiness.	System behavior cannot be mistaken for analytical meaning.

Foundation
Type, spacing, state, focus, responsive rules.

>

Component
Metric, chart, table, filter, navigation, feedback.

>

Pattern
Overview, compare, inspect, recover, review.

>

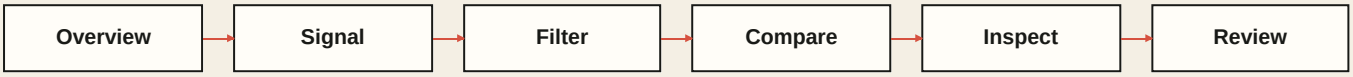
Outcome
Meaning remains stable across views and states.

A design system for analytics is a library of reliable meaning.

DELIVER

The complete solution connects six stages as one analytical system.

The final experience does not end at a polished dashboard. It connects orientation, investigation, evidence, and review through persistent context and reusable behavior.



01 / Overview and signal

- Portfolio, benchmark, period, and freshness establish context.
- One dominant signal identifies where attention should go.
- Status uses text and shape as well as color.

02 / Filter and compare

- Controls sit beside the data they change.
- Active scope and comparison entities remain visible and stable.
- Tables provide precision; charts reveal relationships.

03 / Inspect evidence

- The parent signal persists through drill-down.
- Supporting drivers appear progressively.
- Definitions, breadcrumbs, and prior filters preserve context.

04 / Review and communicate

- Freshness, definitions, assumptions, and completeness are checked.
- Loading, empty, error, success, and review-readiness states are explicit.
- The result can be explained without reconstructing the workflow.

Final screen inventory

ID	SCREEN	PRIMARY JOB	KEY CONTEXT RETAINED
FS-01	Portfolio overview	Identify the signal requiring attention.	Portfolio, benchmark, period, freshness.
FS-02	Signal detail	Understand what changed and why it matters.	Definition, variance, status, parent overview.
FS-03	Filter and compare	Test explanations against stable entities.	Selected entities, units, dates, applied scope.
FS-04	Driver inspection	Open supporting evidence without losing orientation.	Parent signal, breadcrumb, prior filters.
FS-05	Review readiness	Verify the result before communication.	Definitions, freshness, assumptions, completeness.
FS-06	System states	Explain loading, empty, error, success, and recovery.	Affected scope and available next action.

DELIVER

The verified result is operational relevance at institutional scale.

The dashboard work supported institutional users responsible for managing institutional assets. The representative study adds directional usability evidence across 12 interviews, 184 survey responses, and three rounds of five participants; no investment-performance outcome is claimed.

What the work delivered

- A six-stage investigation from overview to review.
- Signal-led hierarchy rather than equal-weight module density.
- Stable comparison context, with controls attached to affected evidence.
- Context-preserving drill-down and explicit recovery.
- Reusable patterns for metrics, charts, tables, filters, navigation, feedback, and review.
- Keyboard, focus, semantic, contrast, and non-color requirements integrated into the system.

Institutional asset context - the documented scope of the workflows supported.

Metric boundary

- **Platform context:** manages institutional assets.
- **Study evidence:** 12 interviews and 184 survey responses.
- **Usability evidence:** three moderated rounds of five participants.
- **Final round:** 5/5 completed the core flow; 4/5 completed the keyboard path unaided.
- **Not claimed:** portfolio performance impact.

RESEARCH SIGNAL	DESIGN RESPONSE	SUPPORTED OUTCOME	STATUS
Weak orientation	Dominant signal hierarchy.	A clearer starting point for investigation.	Validated qualitatively; no task-time claim.
Unstable context	Persistent portfolio, benchmark, dates, units, and filters.	More defensible like-for-like comparison.	Validated through comparison and scope tasks.
Context loss	Parent signal, breadcrumb, and retained scope.	A continuous investigation path.	5/5 completed the final core flow.
Ambiguous system state	Explicit loading, empty, error, success, and readiness patterns.	System behavior is less likely to be mistaken for data meaning.	5/5 interpreted status without color alone.
Institutional scale	Responsive dashboards and reusable analytical patterns.	Operational relevance for teams managing institutional assets.	Verified project context.

Research sources and reconstruction notes

External sources support the design rationale. Study counts, quotations, survey totals, and usability results are representative, anonymized reconstructions authorized for this portfolio narrative; no client identities or portfolio values are shown. The institutional asset-management context remains documented project scope.

[1] Santoro, F. de O., Oliveira, A. R. de, & Silva Júnior, D. S. da (2024). Systematic review of chart formats in administrative and financial dashboards. DOI: 10.56238/arev6n4-256.
https://doi.org/10.56238/arev6n4-256

[2] Liu, Y., & Shen, Z. (2025). Dashboard design form and information-reading performance under time pressure. *Frontiers in Psychology, 16*, 1635951. DOI: 10.3389/fpsyg.2025.1635951.
https://doi.org/10.3389/fpsyg.2025.1635951

[3] Almasi, S., Bahaadinbeigy, K., Ahmadi, H., Sohrabei, S., & Rabiei, R. (2023). Usability Evaluation of Dashboards: A Systematic Literature Review of Tools. DOI: 10.1155/2023/9990933.
https://doi.org/10.1155/2023/9990933
https://pearl.plymouth.ac.uk/nm-research/587/

[4] BlackRock Aladdin. Public product material on integrated risk, return, scenario analysis, asset allocation, dashboards, and data access.
https://www.blackrock.com/institutions/en-us/investment-capabilities/technolgy/aladdin-portfolio-management-software

[5] FactSet. Portfolio Analysis product brochure describing multi-asset performance, attribution, risk, reporting, exposure, and stress testing.
https://insight.factset.com/hubfs/Resources%20Section/Brochures/portfolio-analysis-brochure.pdf?t=1521035901234

[6] Morningstar Direct. Public product material on investment discovery, screening, portfolio composition, stress testing, attribution, dashboards, and reporting.
https://www.morningstar.com/business/brands/data-analytics/products/direct?elqcampaignid&prd=Research

How the evidence was used

- **Chart and hierarchy decisions:** [1], [2].
- **Task and usability criteria:** [3].
- **Competitive capability baseline:** [4], [5], [6].
- **Project decisions:** surviving Fundscope workflow, wireframe, evaluation, system, and outcome record.

Evidence boundary: participant counts, quotations, survey totals, and usability results are representative research synthesis for this anonymized portfolio narrative. The institutional asset-management context is documented project scope; no investment-performance outcome is claimed.